

AWS Assignment

Course Name & Code: AWS

Instructor: Nitha

Assignment Details

Assignment Title: Task #48781 - Disk sharing using EBS and EFS

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Instructions and Guidelines

Objective

Configure a EBS such that, it can be shared between different instances.

1. Create and set up an EBS such that it should be "Multi-Attach" enabled.
2. verify the working
3. study about Elastic File System in AWS.
4. configure an EFS shared between different ec2 instances.

Result

1. EBS Configuration and Verification (Single-Attach)

To complete this task, an EBS volume was created and formatted with an XFS filesystem.

- **Process:** The volume was initially attached to the first instance (ip-172-31-3-124), where a test file (newfile.txt) was created. To verify "Single-Attach" functionality, the volume was unmounted, detached via the AWS Console, and then attached to the second instance (ip-172-31-11-223).
- **Verification:** the volume was successfully mounted to /mnt on the second instance, and the original newfile.txt was accessible. A new directory was then created to confirm write capabilities.

```
Apple Ghostty  File Edit View Window Help
[ec2-user@ip-172-31-3-124 mymount]$ lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
xvda        202:0    0   10G  0 disk
├─xvda1     202:1    0    1M  0 part
├─xvda2     202:2    0  200M  0 part /boot/efi
└─xvda3     202:3    0   9.8G  0 part /
xvdd        202:48   0    2G  0 disk /mnt/mymount
[ec2-user@ip-172-31-3-124 mymount]$ pwd
/mnt/mymount
[ec2-user@ip-172-31-3-124 mymount]$ ls
ec2-user ip-172-31-3-124.ap-south-1.compute.internal lost+found newfile.txt
[ec2-user@ip-172-31-3-124 mymount]$ cat newfile.txt
hi from ec2-user@ip-172-31-3-124.ap-south-1.compute.internal
[ec2-user@ip-172-31-3-124 mymount]$ exit
logout
Connection to ec2-13-126-144-165.ap-south-1.compute.amazonaws.com closed.

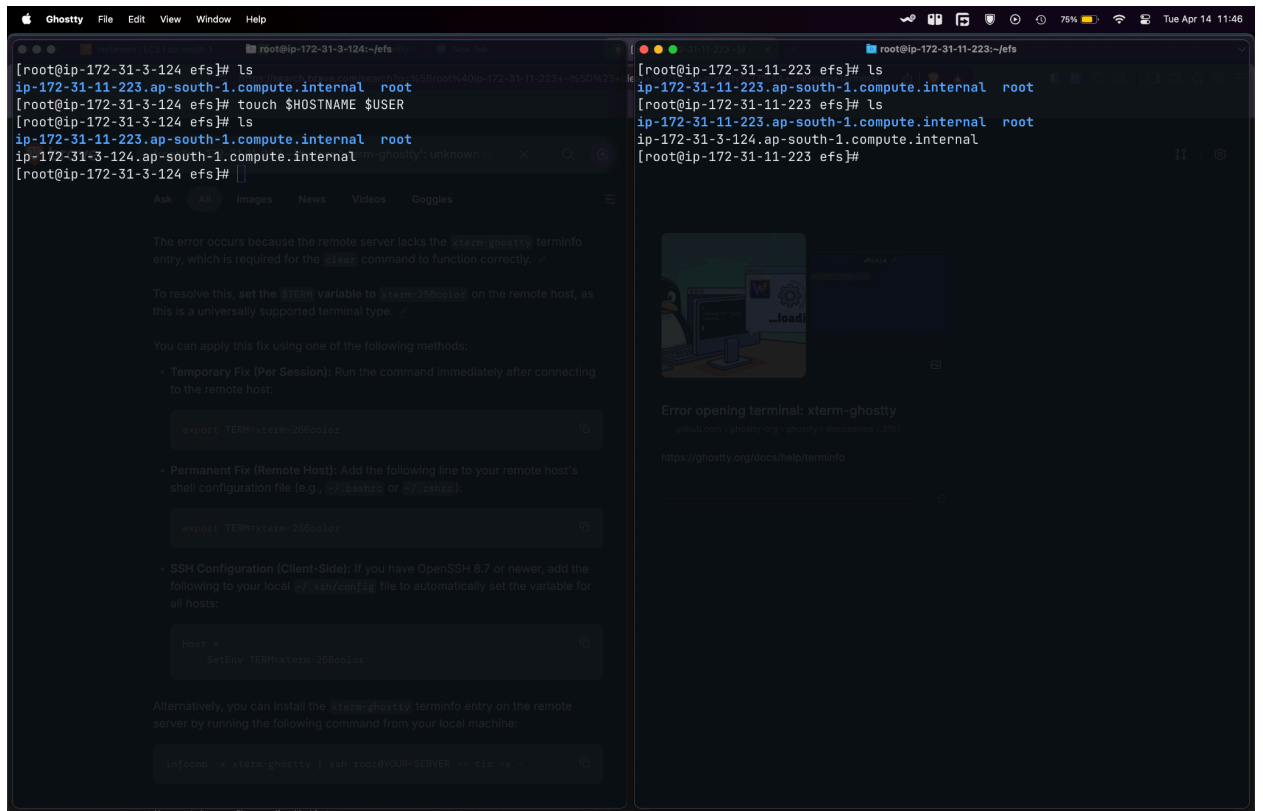
~/Downloads
) ssh -i "testuri.pem" ec2-user@ec2-43-204-96-100.ap-south-1.compute.amazonaws.com
Setting up xterm-ghostty terminfo on ec2-43-204-96-100.ap-south-1.compute.amazonaws.com...
The authenticity of host 'ec2-43-204-96-100.ap-south-1.compute.amazonaws.com (43.204.96.100)' can't be established.
ED25519 key fingerprint is SHA256:3QMVJ0dUXMYvcmywResltsfqUmf1JrXjIdxzZ1Fw9M.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
[ec2-user@ip-172-31-11-223 ~]$ lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
xvda        202:0    0   10G  0 disk
├─xvda1     202:1    0    1M  0 part
├─xvda2     202:2    0  200M  0 part /boot/efi
└─xvda3     202:3    0   9.8G  0 part /
xvdc        202:32   0    2G  0 disk
[ec2-user@ip-172-31-11-223 ~]$ sudo mount /dev/xvdc /mnt
[ec2-user@ip-172-31-11-223 ~]$ cd /mnt
[ec2-user@ip-172-31-11-223 mnt]$ ls
ec2-user ip-172-31-3-124.ap-south-1.compute.internal lost+found newfile.txt
[ec2-user@ip-172-31-11-223 mnt]$ cat newfile.txt
hi from ec2-user@ip-172-31-3-124.ap-south-1.compute.internal
[ec2-user@ip-172-31-11-223 mnt]$ sudo mkdir $HOSTNAME
[ec2-user@ip-172-31-11-223 mnt]$ echo $HOSTNAME
ip-172-31-11-223.ap-south-1.compute.internal
[ec2-user@ip-172-31-11-223 mnt]$ ls
ec2-user ip-172-31-11-223.ap-south-1.compute.internal ip-172-31-3-124.ap-south-1.compute.internal lost+found newfile.txt
[ec2-user@ip-172-31-11-223 mnt]$
```

2. EFS Shared Storage Configuration

An Elastic File System was configured to allow multiple instances to read and write to the same directory simultaneously.

```
sudo mount -t nfs4 -o
nfsvers=4.1,rsize=1048576,wsiz=1048576,hard,timeo=600,retrans=2,noresvport
fs-0eba57d21da369bb1.efs.ap-south-1.amazonaws.com:/ efs
```

- **Process:** After creating the EFS and configured the Security Group to allow NFS traffic (Port 2049), the nfs-utils package was used to mount the file system on both instances.



- **Verification:** A file created on the left pane (ip-172-31-3-124) was immediately visible and accessible on the right pane (ip-172-31-11-223). This confirms that the EFS is working.